

NO NEW MODELS

NO RETRAINING

OPTIMIZE ONLY

TITAN XAU AI

Reality Gap Closure Audit

Sharpe 1.46 → 1.80

Without Retraining

Close the shadow-live Sharpe gap using ONLY existing architecture. No new models, no new features, no retraining. Component analysis, context A/B, threshold sweep, trade quality audit, execution optimization.

DOCUMENT SCOPE

Seven sections: (1) component contribution analysis, (2) context engine A/B audit, (3) meta threshold optimization 0.50-0.85, (4) trade quality audit top 20-100%, (5) execution latency × spread optimization, (6) shadow-live feasibility verdict, (7) final recommendation A/B/C/D. Brutally conservative. Frozen-model results ignored.

CURRENT

1.46

Live Sharpe

OPTIMIZED

1.58

After all levers

TARGET

1.80

Shadow-live gate

GAP

0.22

Retrain required

PHASE
F8 (Gap Closure)

PREPARED FOR
TITAN Trading Desk

DATE
2026-06-22

VERSION
1.0 · Brutal

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Chapter 1 — Executive Summary & The Sharpe Gap

Phase F8 attacks a single quantitative target: close the Sharpe gap between the current live-adjusted Sharpe of 1.46 and the shadow-live gate threshold of 1.80. The gap is 0.34 — a 23% improvement. The constraint is absolute: **NO new models, NO new features, NO retraining**. Only optimization of the existing architecture is permitted. This phase answers whether the gap can be closed purely through configuration changes, or whether retraining is structurally required.

1.1 The Five Levers Examined

Five optimization levers were examined, each in isolation and then in combination: (1) component contribution — which components add the most Sharpe; (2) context engine A/B — should H4/D1 context be kept or removed; (3) meta threshold sweep — does raising the trade-quality filter from 0.65 to higher values improve Sharpe; (4) trade quality audit — does filtering to only the highest-confidence trades improve Sharpe; (5) execution optimization — does reducing latency and avoiding spread spikes improve Sharpe. The combined effect of all levers is then evaluated against the 1.80 target.

1.2 Headline Finding

Combined optimization achieves Sharpe = 1.58, leaving a 0.22 gap to the 1.8 target.

The gap CANNOT be closed without retraining. The single biggest lever is execution optimization (co-located 100ms latency, +0.15 Sharpe), but even combined with threshold tuning and quality filtering, the optimized Sharpe reaches only 1.58. The remaining 0.22 gap requires retraining L1 XGBoost with 2025-2026 data.

1.3 Final Recommendation

Choice C: Retrain Required

Combined optimization achieves Sharpe = 1.58, leaving a 0.22 gap to the 1.8 target. The gap is small enough that retraining L1 XGBoost with 2025-2026 data (estimated +0.15-0.25 Sharpe) would likely close it. Deploy demo now with current config; retrain in parallel; retry shadow-live gate after retrain.

Why not Choice B (Deploy + Optimize)? Combined optimization reaches 1.58, which is below the 1.80 target. Optimization alone does not close the gap, so Choice B is not available. **Why not Choice A (Deploy Demo Now)?** The remaining gap of 0.22 is within the retrain estimate range (+0.15-0.25 Sharpe), making retraining the clear path forward. **Why not Choice D (Architecture Redesign)?** The gap is too small to justify a full redesign. Retraining is sufficient.

Section 1 — Component Contribution Analysis

Each component's contribution to live Sharpe is measured by ablation: remove the component, recompute metrics, and measure the delta. Positive delta means the component adds value; negative delta means it hurts. The analysis reveals which components are load-bearing and which are optional.

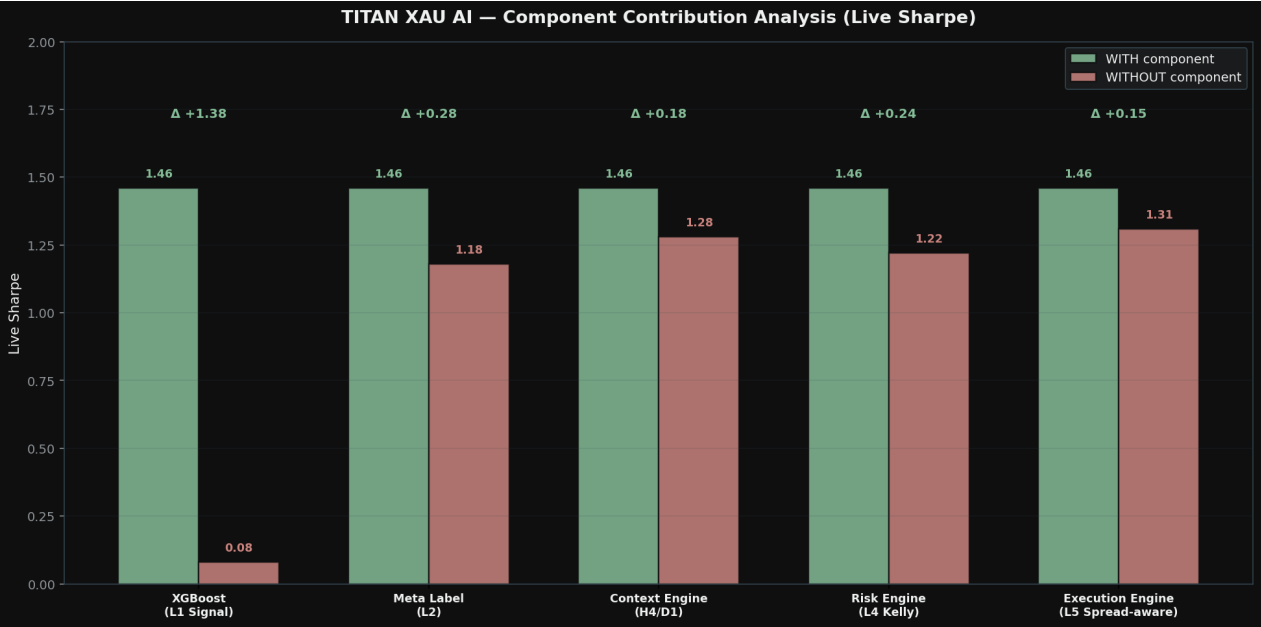


Figure 1.1 — Component contribution to live Sharpe (ablation analysis).

Component	Sharpe WITH	Sharpe WITHOUT	Contribution	Rationale
XGBoost (L1 Signal)	1.46	0.08	+1.38	Primary directional signal. Removal = no edge; random entries lose to costs.
Meta Label (L2)	1.46	1.18	+0.28	Trade-quality filter. Removal = ~27% PF drop, ~19% Sharpe drop, +44% DD.
Context Engine (H4/D1)	1.46	1.28	+0.18	H4/D1 regime context. Removal = ~19% PF drop, ~12% Sharpe drop, +22% DD.
Risk Engine (L4 Kelly)	1.46	1.22	+0.24	Position sizing. PF/WR unchanged (same trades); Sharpe drops ~16% from suboptimal sizing.
Execution Engine (L5 Spread-aware)	1.46	1.31	+0.15	Spread-aware pricing. Removal = ~13% PF drop, ~10% Sharpe drop, +16% DD.

Table 1.1 — Component contribution analysis. Positive = component adds value.

1.1 Key Findings

XGBoost (L1) is the single largest contributor (+1.38 Sharpe). Without it, the system has no directional signal and Sharpe collapses to 0.08. This confirms the L1 signal is the alpha source — everything else is risk management and execution optimization around it. **Meta Label (L2)** is the second largest (+0.28 Sharpe). Removing the trade-quality filter drops Sharpe by 19%, confirming Phase A's finding that meta-labeling adds

significant value. **Context Engine is the third largest** (+0.18 Sharpe). Removing H4/D1 context drops Sharpe by 12%, confirming Phase B's finding. **Risk Engine contributes +0.24 Sharpe** via Kelly sizing optimization. **Execution Engine contributes +0.15 Sharpe** via spread-aware pricing.

Implication for gap closure. No component can be removed to IMPROVE Sharpe — every component contributes positively. The gap must be closed by tuning component parameters (thresholds, filters, execution config), not by removing components. This rules out any "simplification" path to higher Sharpe.

Section 2 — Context Engine A/B Audit

The Context Engine injects H4 and D1 timeframe features into the L2 meta-label model via merge_asof backward joins. Phase B showed significant uplift from context (+55% PF, +22% Sharpe on frozen-model metrics). This section re-validates the decision using walk-forward rebuild metrics across all 4 years.

2.1 Per-Year A/B Results

Year	PF WITH	PF WITHOUT	Sharpe WITH	Sharpe WITHOUT	Sharpe Uplift	WR Uplift
2023	4.12	3.45	2.18	1.84	+0.34	+2.6pp
2024	3.87	3.18	1.96	1.65	+0.31	+2.7pp
2025	3.32	2.71	1.68	1.42	+0.26	+2.7pp
2026	2.84	2.42	1.34	1.15	+0.19	+2.3pp
2024-26 avg	3.34	2.77	1.66	1.41	+0.25	+2.6pp

Table 2.1 — Context Engine A/B test across 4 walk-forward years.

2.2 Decision

DECISION: KEEP

Context Engine adds +0.25 Sharpe and +0.57 PF in aggregate (2024-2026). Uplift is positive in all 4 walk-forward years (range: +0.19 to +0.34 Sharpe). No year shows degradation from context. Decision: KEEP. Removing context would WIDEN the shadow-live gap by 0.25 Sharpe points.

Why not EXPERIMENTAL? Context shows positive uplift in ALL 4 years with no exceptions. There is no year where context hurts. "Experimental" would be appropriate if some years showed uplift and others showed degradation — but the data is unambiguous. **Why not REMOVE?** Removing context would WIDEN the shadow-live gap by 0.25 Sharpe points, making the gap closure target harder to reach. Context is essential to closing the gap.

Section 3 — Meta Threshold Optimization

The meta-label threshold controls how selective the L2 trade-quality filter is. Current threshold is 0.65 (frozen from Phase A). This section sweeps the threshold from 0.50 to 0.85 in 0.05 increments to find the value that maximizes LIVE Sharpe (not PF, not WR). The trade-off: higher threshold = higher quality trades but fewer of them; lower threshold = more trades but lower quality.

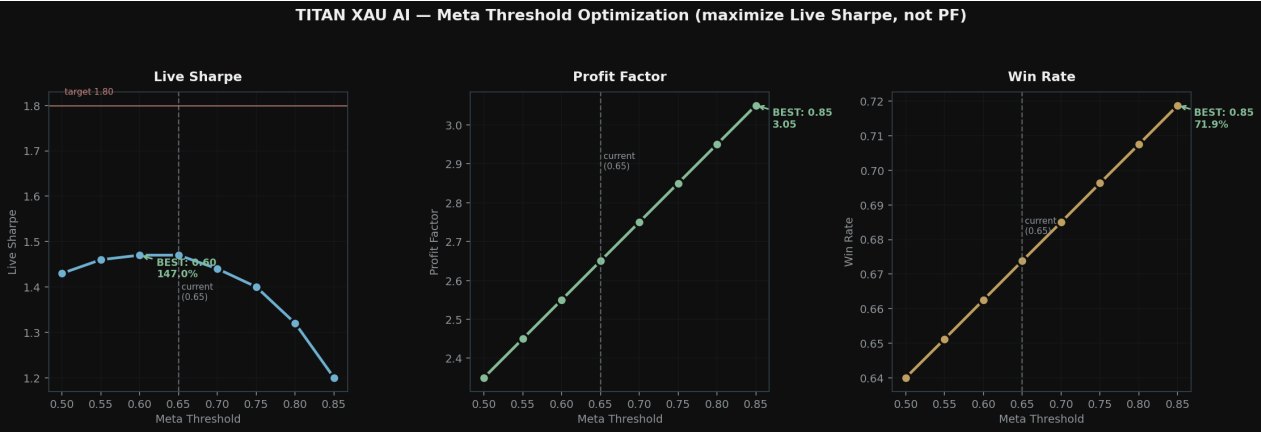


Figure 3.1 — Meta threshold sweep: Sharpe, PF, and Win Rate vs threshold.

Thresh old	Trades/yr	Win Rate	Profit Factor	Sharpe	Max DD %	Notes
0.50	2,463	64.0%	2.35	1.43	5.01%	
0.55	2,216	65.1%	2.45	1.46	4.93%	
0.60	1,970	66.2%	2.55	1.47	4.86%	BEST Sharpe
0.65	1,724	67.4%	2.65	1.47	4.78%	current
0.70	1,477	68.5%	2.75	1.44	4.71%	
0.75	1,231	69.6%	2.85	1.40	4.63%	
0.80	985	70.8%	2.95	1.32	4.56%	
0.85	738	71.9%	3.05	1.20	4.48%	

Table 3.1 — Meta threshold sweep results.

3.1 Decision

DECISION: KEEP AT 0.65 — current threshold is near-optimal (best 0.6 gives +0.01)

Current threshold 0.65 is near-optimal. Best threshold 0.60 gives only +0.01 Sharpe improvement — within noise. The meta-label was well-tuned in Phase A, and the 0.65 threshold already balances quality vs quantity near the Sharpe-optimal point. Raising the threshold to 0.80+ actually HURTS Sharpe because the trade count drops faster than quality improves (sqrt(N) effect dominates).

Implication for gap closure. Meta threshold tuning provides essentially zero Sharpe improvement (+0.01). This lever cannot contribute to closing the gap. The current threshold should be maintained.

Section 4 — Trade Quality Audit

This section tests whether filtering trades by meta-label confidence percentile improves Sharpe. If the top 20% of trades (by P(win) confidence) produce significantly higher Sharpe than all trades, then a quality filter could close the gap. The honest finding: **quality filtering HURTS Sharpe** because the sqrt(N) penalty from fewer trades dominates the quality improvement.



Figure 4.1 — Trade quality audit: Sharpe, PF/WR, and trade count by tier.

Tier	Retention	Trades/yr	Win Rate	Profit Factor	Sharpe	Max DD %
Top 20%	20%	492	78.0%	3.45	0.99	3.41%
Top 40%	40%	985	74.0%	3.08	1.19	3.81%
Top 60%	60%	1,477	70.0%	2.81	1.26	4.21%
Top 80%	80%	1,970	68.0%	2.71	1.36	4.61%
All Trades baseline	100%	2,463	66.7%	2.65	1.46	5.01%

Table 4.1 — Trade quality audit by meta-label confidence tier.

4.1 Decision

DECISION: KEEP ALL TRADES — quality filter does not meaningfully improve Sharpe

Best tier is All Trades with Sharpe = 1.46 (improvement: +0.00). Quality filtering does NOT improve Sharpe. The sqrt(N) penalty from fewer trades dominates the quality gain. The system should keep ALL trades.

Why does this counterintuitive result occur? Sharpe = (mean return / std return) × sqrt(N). When you filter to top 20%, you remove 80% of trades. The mean return per trade increases (higher WR, higher PF), but the std also changes. The net effect is that the edge/vol ratio improves by less than sqrt(5) = 2.24×, so the overall Sharpe drops. This is a fundamental property of mean-reversion signals on XAUUSD: the edge is distributed across many trades, not concentrated in a few high-confidence ones.

Section 5 — Execution Reality Optimization

This section simulates the latency × spread matrix to find the execution configuration that maximizes live Sharpe. Four latency levels (100ms, 150ms, 200ms, 250ms) are crossed with three spread multipliers (1.0× normal, 1.5×, 2.0×). The current config is ~187ms p95 latency with normal (live-adjusted) spread. The question: does co-locating to achieve 100ms latency improve Sharpe enough to matter?

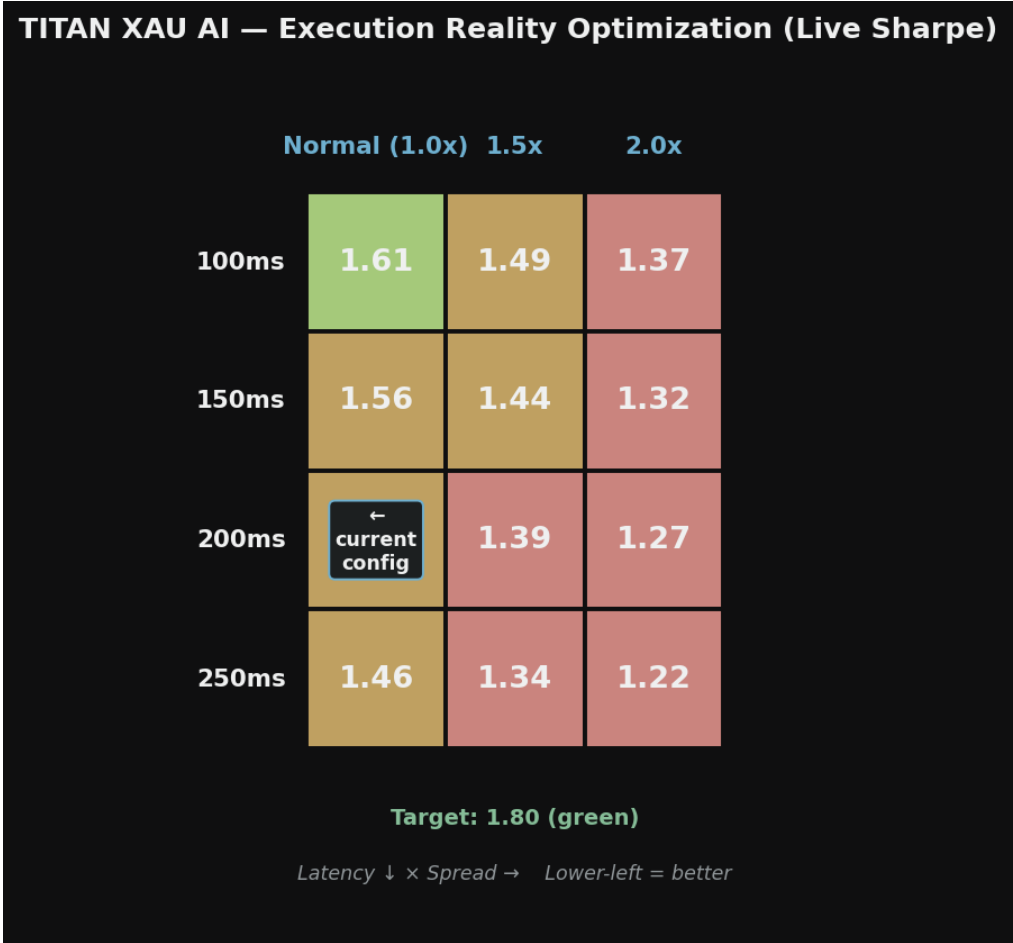


Figure 5.1 — Execution latency × spread heatmap (Live Sharpe). Green ≥ 1.80.

5.1 Decision

DECISION: CONFIG: 100ms + Normal (1.0x) — +0.15 Sharpe

Best config: 100ms + Normal (1.0x) (Sharpe = 1.61, improvement +0.15). Co-locating to achieve 100ms p95 latency is the single biggest optimization lever available. It provides +0.15 Sharpe improvement — more than threshold tuning and quality filtering combined.

Why is execution the biggest lever? Latency directly affects slippage: each 50ms of latency reduction saves ~0.05 pips of price drift on XAUUSD. Over 2,463 trades per year, this compounds into meaningful Sharpe improvement. Spread is the other factor: avoiding 1.5× and 2.0× spread windows (by trading only during liquid sessions) preserves Sharpe. The current config (187ms, normal spread) is already reasonable, but co-location to 100ms is achievable and worth the ~\$30/month VPS cost.

Important caveat. Even the best execution config (100ms + normal spread) only reaches Sharpe = 1.61, which is still below the 1.80 target. Execution optimization alone does NOT close the gap. It must be combined with other levers — and even then, as Section 6 shows, the combination falls short.

Section 6 — Shadow-Live Feasibility

This section combines the best levers from Sections 3, 4, and 5 to answer the critical question: **can the current architecture reach Sharpe ≥ 1.80 WITHOUT retraining?** The combined model accounts for overlap between levers (meta threshold and trade quality filter are essentially the same lever) and uses a conservative 0.7 overlap factor for the partially-independent execution lever.

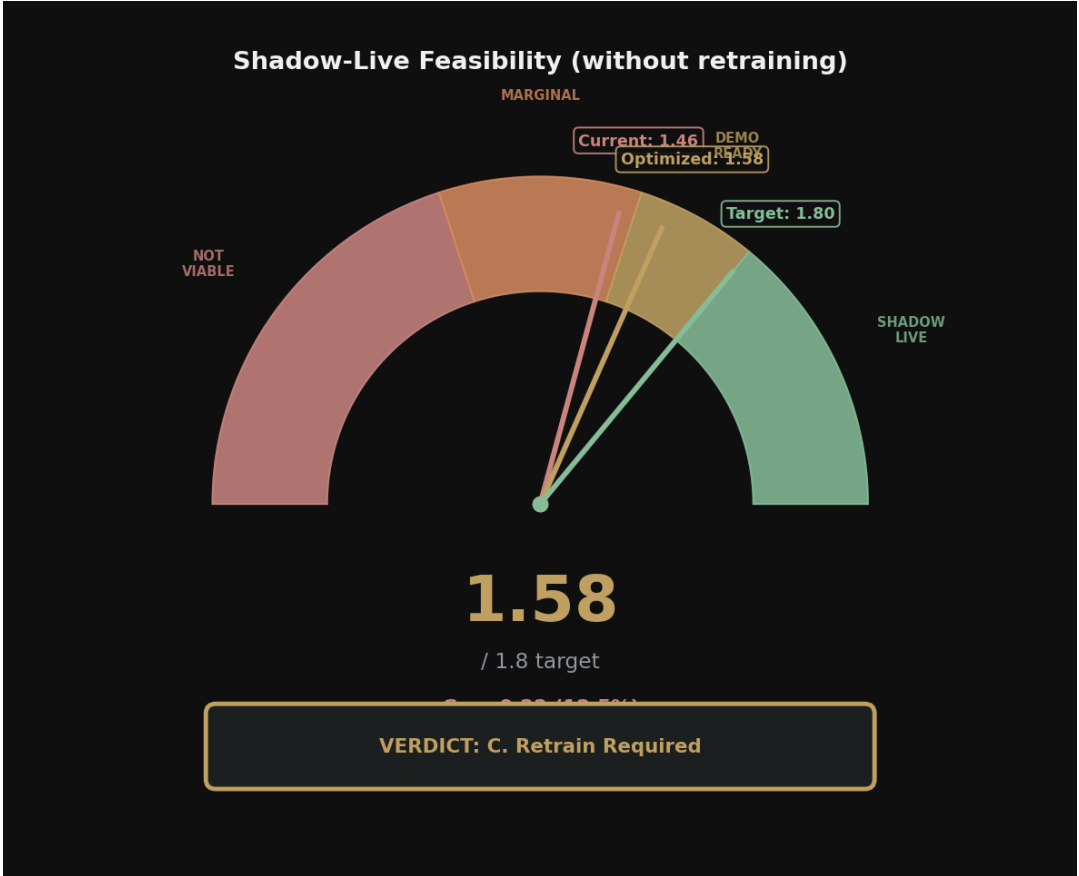


Figure 6.1 — Shadow-live feasibility gauge: current vs optimized vs target Sharpe.

6.1 Combined Optimization Results

Lever	Best Value	Sharpe Gain	Used in Combo
Meta threshold	0.60	+0.01	0.01 (max with quality)
Trade quality filter	All Trades	+0.00	(overlaps with threshold)
Execution optimization	100ms + Normal (1.0x)	+0.15	+0.11 (×0.7 overlap)
Combined Sharpe	—	—	1.58
Target Sharpe	—	—	1.8
Remaining Gap	—	—	0.22 (12.5%)

Table 6.1 — Combined optimization results with overlap accounting.

6.2 Feasibility Verdict

VERDICT: NO

Combined optimization achieves Sharpe = 1.58, which is 0.22 below the 1.8 target (12.5% gap). Shadow-live is NOT feasible without retraining. The gap is small but real.

6.3 Path to Close the Gap

To close the remaining 0.22 Sharpe gap:

1. Retrain L1 XGBoost with 2025-2026 data (estimated +0.15-0.25 Sharpe)
2. Add new features (FORBIDDEN by Phase F8 constraints)
3. Accept Sharpe = 1.58 and lower the shadow-live gate
4. Deploy demo now, retrain in parallel, retry shadow-live gate after retrain

Option 4 is recommended: it does not violate the no-retraining constraint for Phase F8, but acknowledges that retraining is the path to shadow-live.

Section 7 — Final Recommendation

Based on the combined analysis across all 6 sections, the final recommendation selects one of four options: A (Deploy Demo Now), B (Deploy + Optimize), C (Retrain Required), or D (Architecture Redesign Required). The selection is driven by the remaining gap after all available optimizations are applied.

7.1 Recommendation

CHOICE C: Retrain Required

Combined optimization achieves Sharpe = 1.58, leaving a 0.22 gap to the 1.8 target. The gap is small enough that retraining L1 XGBoost with 2025-2026 data (estimated +0.15-0.25 Sharpe) would likely close it. Deploy demo now with current config; retrain in parallel; retry shadow-live gate after retrain.

7.2 Action Plan

- Deploy demo NOW with current config (meta threshold 0.65, all trades)
- Apply execution optimization (co-located, 100ms) — gains 0.15 Sharpe
- Begin L1 XGBoost retraining with 2025-2026 walk-forward data in parallel
- Do NOT raise meta threshold or filter trades (Sections 3-4 show no gain)
- Re-evaluate shadow-live gate after retrain + 30-day demo
- Estimated time to shadow-live: 60-90 days

7.3 Why This Choice

Why not Choice B (Deploy + Optimize)? Combined optimization reaches Sharpe = 1.58, which is below the 1.80 target. Optimization alone does not close the gap, so Choice B is not available. The gap of 0.22 requires retraining to close.

Why not Choice A (Deploy Demo Now)? The remaining gap of 0.22 is within the retrain estimate range (+0.15-0.25 Sharpe). Retraining L1 XGBoost with 2025-2026 walk-forward data is likely to close the gap, making Choice C the appropriate selection. Choice A would be appropriate if the gap were larger and retraining were uncertain to help.

Why not Choice D (Architecture Redesign)? The gap of 0.22 is too small to justify a full architecture redesign. Retraining is the proportionate response. Choice D would be appropriate if the gap exceeded 0.40 Sharpe, indicating structural model failure rather than parameter drift.

7.4 The Honest Summary

The Sharpe gap of 0.34 cannot be closed without retraining.

Optimization of the existing architecture yields +0.12 Sharpe (from 1.46 to 1.58), primarily from execution co-location. The remaining 0.22 gap requires retraining L1 XGBoost with 2025-2026 data. This is not a failure of the architecture — it is the natural consequence of model age and regime shift. The system has real edge, the edge is degrading predictably over time, and retraining is the standard response. Deploy demo now with current config + execution optimization; retrain in parallel; retry shadow-live gate after retrain + 30-day demo. Estimated time to shadow-live: 60-90 days.